

# 3D Stokes Polarimetry of Dipolar Near-Fields

Guillermo Serrera<sup>1,\*</sup>, Yael Gutiérrez<sup>1</sup>, José J. Gil<sup>2</sup>, Fernando Moreno<sup>1</sup>

<sup>1</sup>Applied Physics Department, University of Cantabria, Spain

<sup>2</sup>Independent researcher, 29620 Casares Costa, Spain

\*Corresponding author: [guillermo.serrera@unican.es](mailto:guillermo.serrera@unican.es)

**Abstract**—Dipolar emitters constitute fundamental building blocks in nanophotonics, yet their polarization behavior outside the far-field approximation remains insufficiently characterized. In this work, we introduce the three-dimensional Stokes formalism as a unified framework to describe the complete polarimetric landscape of dipolar radiation across the near-, intermediate-, and far-field regimes.

**Keywords**— Stokes, polarimetry, dipolar fields.

## I. INTRODUCTION

Dipolar emitters play a central role in nanoscale light–matter interactions, yet their polarization properties remain insufficiently characterized outside the far-field approximation. Conventional 2D Stokes analysis assumes purely transverse fields and therefore cannot capture the intrinsically three-dimensional, vectorial nature of dipolar radiation in the near- and intermediate-field regimes. Recent experimental advances have further underscored this limitation by demonstrating the measurement of full 3D polarization and optical spin in evanescent and near-field environments.[1] A rigorous description of these regimes is therefore essential for nanophotonic applications, where strongly inhomogeneous electromagnetic fields are ubiquitous.

In this work, we apply the 3D Stokes formalism to analyze the emission of a range of dipolar sources, including electric, magnetic, spinning, and chiral dipoles. We show how this formalism effectively describes all polarization phenomenology in real-space, in contrast with approaches based on the Angular Spectrum Representation.[2] Furthermore, we show how, for deterministic fields, the 3D Stokes formalism allows simplification to 2D representations, accommodating the interpretation of complex polarization landscapes to conventional polarimetric techniques.

## II. METHODS

Our analysis is based on the 3D polarization matrix constructed from the full vectorial electric field of dipolar emitters. The matrix is expanded using the Gell–Mann basis, yielding nine real-valued 3D Stokes parameters.[3] These parameters encode total intensity, linear and circular polarization content, and the relative weight of radial versus transverse field components.

## III. RESULTS AND DISCUSSION

For linear electric dipoles, the 3D Stokes formalism reveals a fully linear polarization in the near field, dominated by longitudinal components. In the intermediate field, non-zero

circular Stokes parameters emerge despite uniform intensity, evidencing transverse optical spin in real space.[2] In the far field, the field becomes purely transverse and is well described by conventional 2D Stokes parameters.

Spinning electric dipoles exhibit uniform azimuthal symmetry and strong circular polarization signatures, with the formalism clearly capturing helicity inversion and the evolution of spin components across regimes.[4]

For chiral dipoles, the near field closely resembles that of an electric dipole, while magnetic contributions become significant in the intermediate regime. The 3D Stokes parameters distinguish chiral circular polarization from electric-dipole-induced polarization artifacts, highlighting potential pitfalls in chiral sensing.

Finally, we show how a reduction of 3D-to-2D polarization states can be achieved by finding an intrinsic system for each point in space. This allows to express the polarization state in terms of the usual 2D formalism, revealing the fundamental nature of evanescent linear and circular polarization components with respect to the energy flow.

## IV. CONCLUSION

We have demonstrated that the 3D Stokes formalism provides a compact, intuitive, and fully analytical framework for characterizing the polarization structure of dipolar emission across near-, intermediate-, and far-field regimes. By explicitly accounting for longitudinal and transverse field components, the formalism captures polarization effects that are inaccessible to conventional 2D Stokes analysis. This approach offers a powerful and accessible alternative to angular-spectrum-based methods and is well suited for the analysis of nanoscale light–matter interactions involving inhomogeneous fields.

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